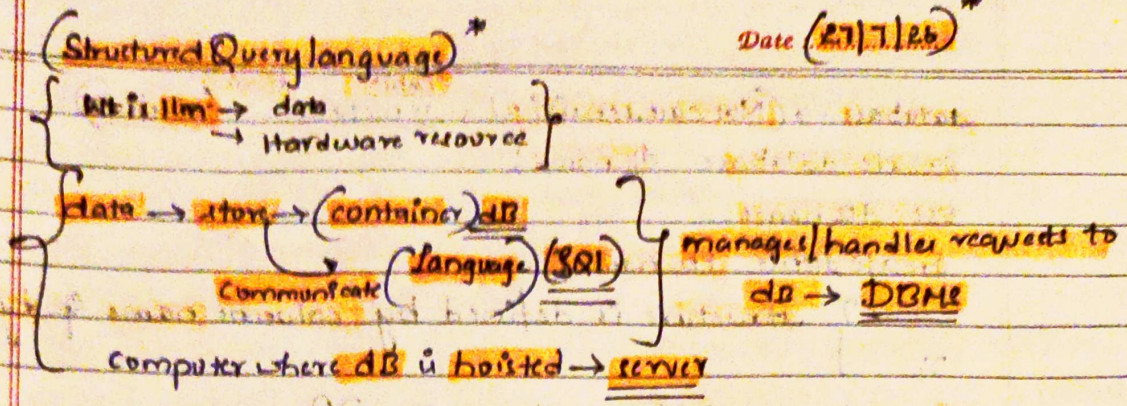


Date (27/7/25)*



Database

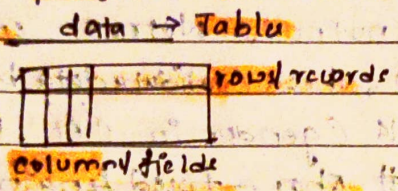
(SQL)

(Not only SQL)

(NoSQL)

① Relational Db
 (MySQL, PostgreSQL, Microsoft SQL)

① Key value storage (DynamoDB)
 ② column based (Cassandra)
 ③ graph based (Neo4j, Amazon Neptune)
 ④ document based (MongoDB)

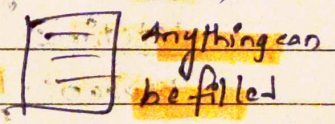
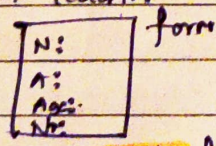


- structured, fixed schema (columns fixed) flexible (fields can vary)
- Vertical scaling (bigger server) Horizontal scaling
- full ACID guarantees (more server)
- Consistency fully (Banking, ERP, Enterprise inventory) (chat analytics, SQL)

Schema:- blueprint of a DB, defines what fields exist, its type of data each field stores & the rules that must follow.

(SQL) -> Teacher

(NoSQL) -> Teacher (Blank)



you cannot add favorite hobby suddenly as there is no box to be filled

SQL Instructions:-

- DDL (Data Definition Language) -> create, Alter, drop, Truncate, rename.
- DQL (Data Query Language) -> (select)
- DML (Data Manipulation Language) -> (Insert, update, delete)
- DCL (Data Control Language) -> Grant, revoke
- TCL (Transaction Control Language) -> (commit, rollback, savepoint)
(manage transactions)

database → (Not case sensitive), ^{Table given by} structure → ^{columns}

create database db_name;

use db_name;

create table tb_name {

} structure is defined by column name & type
};

Q) create db → table → column (type)?

create db Puni;

use puni;

create table EC

studentid int primary key, # integer, uniquely identifies, Null, dupli

firstname varchar (50) not null, # varchar - variable length,

lastname varchar (50) not null, Not null - empty (X)

age int check (age > 18), check (validate data before storing it)

gender char (1) check (gender in ('M', 'F'))

}; # short - fixed length.

Notes

select database (); shows currently selected db (If No → NULL)

use Puni; To verify which db is currently active

(opens up folder) before executing queries.

constraints: rules enforced by db to maintain data integrity.

primary key, foreign key, unique, not null, check, default.

float: 10.8

date: only date

datetime: datetime

boolean: - 0/1

(F T)

Q) Insert value into EC table?

insert into EC values (2, 'dhanush', 'd', 22, 'M')

adds one row to table Notes: values must follow order of col

data type match (✓)

primary key & duplicated (X)

Q1. select?

EC # retrieves all rows & columns,
 * → give me every column.

Q2. Alter?

alter table EC # Adds new column

add email varchar(60);

Q3. Modify?

alter table EC # change definition of existing column

modify email varchar(70); → varchar(60) → varchar(80)

Q4. Rename column?

alter table EC # change only column name, data stays

rename column email to studentmail;

Q5. Drop column?

alter table EC # delete column permanently, data of

drop column studentmail; studentmail is all lost.

Add vs Modify → change existing column name etc.

↓
create new column

Q6. Truncate:-

Truncate table EC; # removes all rows from table, but
 (empty box but box stays) keeps the table itself,
 (columns stay)*

Q7. foreign key? A foreign key is a column that points to primary
 key of another table.

Why? avoids duplication, suppose there are 500 classes
 without foreign keys → john john ... stored
 500 times (class storage), maintains consistency, easier update
 with foreign key 101, 101 ... 101 but teacher info stored
 only once.

- value may be repeated.
 - can be Null cell.

Referential Integrity

(delete on both, error on both)

create table teachers(
 teacherID int primary key,
 firstname varchar(50) not null;
);

create table classes(
 classID primary key,
 teacherID int,
 foreign key (teacherID)
 references teachers
 (teacherID)
);

Q update?

update ec # update an existing record)

where set age = 18

syntax: update tablename

where age studentid = 2;

set column = value

Q Where? (where clause)

where condition)

select * from students # filters based on a condition

where lastname = 'raj'; common operators: =, >, <, >=, <=

Q orderby?

select studentid, age

from students

order by age asc; (sort in asc) # (desc)

Q distinct?

select distinct studentid # removes duplicate values from students, result

21

21

21

→

18

18

21

18

Q Count(*)?

counts the total no of rows

select count (*) → every row from students;

only for
select count (age)
from students

where ile orderby distinct v/s count

(filter row)

(sorts)

(unique)

(how many rows)

Every row gets updated where has been committed marked

Q Views:- (Virtual table) → store result of an sql select query

→ repetition (x), security ✓, abstraction (✓)

create view (student2details) as

select firstname from students

where stid = 2;

→ select * from student2details

updatable view → join, groupby, having, aggregate

distinct (x) in select (update) *

(After & drop view) *

→ up data table

→ Non updatable

→ u don't have to this time again

Join:- way to combine two/more tables based on common or related column. Inner, left, right, outer, cartesian product (cross join)

studentid	firstname	lastname	age	gender	clauID
1	Vamshi	bhavan	13	M	101
2	sarav	sai	15	F	102
3	diti	gaur	14	F	Null

Inner join

clauID	clauName	TeacherID
101	6A	1
102	6B	2
103	6C	Null

Note:-
(-) comments in mysql

Inner join students, clau ^{based on} → clauID (we have this as common in both

(Common ga unna records show chutham) tables)

select s.firstname as studentname, c.clauName
from students s, clau
inner join clau c on s.clauID = c.clauID

o/p:-

studentname	clauName
Vamshi	6A
Saravali	6B

Null(x)

Syntax:- select columns
from table 1
join table 2 on related column conditions.

Left join (or) Right join (depends on order)

Left → Table 1 priority, Right → Table 2 priority.

select s.firstname, c.clauName
from students s

left join clau c on s.clauID = c.clauID Right join

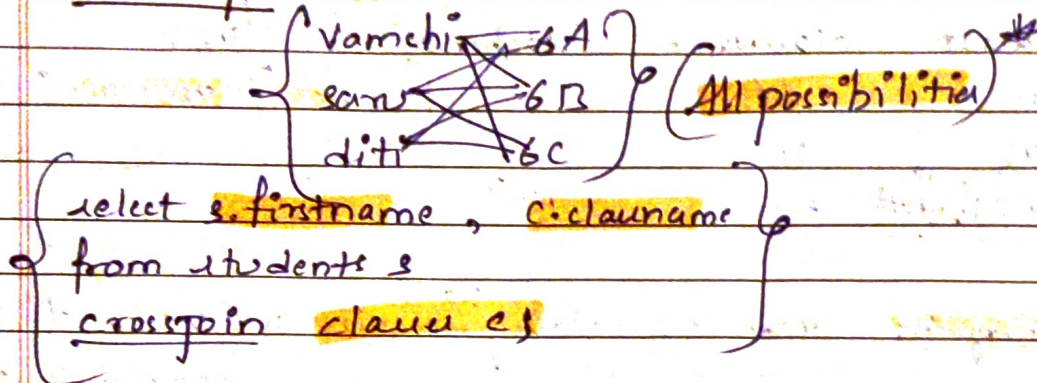
o/p:-	Vamshi	6A	Vamshi	6A
	Sarav	6B	Sarav	6B
	Diti	Null	Diti	6C
			Null	

outer join: Inner join $\rightarrow$ Null (X)
 $\rightarrow$ Null padathadi

(Equal priority untadhi)

Vamshi	101	Vamshi	6A
sarav	102	sarav	6B
diti	Null	diti	Null
		Null	6C

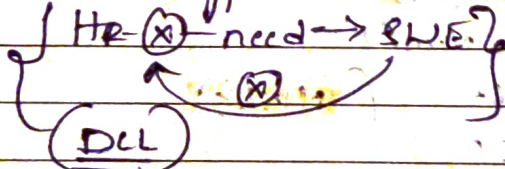
Cartesian product:-



TCL, DCL:-

Idea:- Company $\rightarrow$ (Employees)
 (Sales)

Not every person has to have access of dr



(Table)
 grant select, insert, ~~update~~ on students to sai;
 revoke insert on students from sai;

TCL:- once transaction begins

eg:- (Banks account)

A $\rightarrow$ 1000

B $\rightarrow$ 500

A $\rightarrow$ 500

B $\rightarrow$ 500

begin $\rightarrow$ commit

$\downarrow$ rollback (resume X)

(Concurrency matter)

begin Transaction

update accounts

set bal = bal - 500

where account ID = 1

$\rightarrow$ rollback

update accounts

set bal = bal + 500

where accID = 2

commit;